

blocking maximum block time for resource blocking is only 1 instance of lower priority tasks. The maximum block time can be calculated by first look at the resource that the task uses. then you look at the lower priority tasks that take the same resource, then you take the maximum of all the lower priority. Must also be aware of priority inheritance so for qns2, B6 even though no resource used but becoz T7 can inherit the priority of T5 thus block T6.

SC3103: Embedded Programming

Tutorial 4: PCP/SRP Schedulability

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1. Consider a sporadic real-time task set comprising 10 tasks $\tau_1, \dots, \tau_{10}$, where τ_1 has the highest and τ_{10} has the lowest priority. Suppose there are 3 resources R_1, R_2 and R_3 , shared by these tasks. The usage profile for these resources are as follows ($RAT_{i,j}$ denotes maximum resource access time for resource R_j by task τ_i):

- $RAT_{3,1} = 5$
- $RAT_{5,1} = 3, RAT_{5,3} = 2$
- $RAT_{7,1} = 4, RAT_{7,2} = 5$
- $RAT_{8,2} = 2, RAT_{8,3} = 2$
- $RAT_{10,1} = 1, RAT_{10,2} = 2, RAT_{10,3} = 3$

Any $RAT_{i,j}$ not mentioned above by default has value 0 and it can be assumed that task τ_i does not use resource R_j . Compute the worst-case blocking times B_i for all the 10 tasks under PCP/SRP.

2. Consider a sporadic real-time task set comprising 7 tasks $\tau_1, \dots, \tau_7$, where τ_1 has the highest and τ_7 has the lowest priority. Suppose there are 3 resources R_1, R_2 and R_3 , shared by these tasks. The usage profile for these resources are as follows ($RAT_{i,j}$ denotes maximum resource access time for resource R_j by task τ_i):

- $RAT_{1,1} = 5$
- $RAT_{2,1} = 4, RAT_{2,2} = 3$
- $RAT_{3,1} = 4, RAT_{3,2} = 5$
- $RAT_{5,3} = 2$
- $RAT_{7,3} = 3$

Any $RAT_{i,j}$ not mentioned above by default has value 0 and it can be assumed that task τ_i does not use resource R_j . Compute the worst-case blocking times B_i for all the 7 tasks under PCP/SRP.